Physics Galaxy
Telegram Test Series

JEE Main Level Practice Test-17
for
JEE & NEET Aspirants

Topic : WAVE OPTICS **Time: 75 Min** **Marking +4 –1**

Section - A : MCQs with Single Option Correct

1. Unpolarised light of intensity I_0 is incident on polariser find out intensity of transmitted polarised light :
(A) I_0 (B) $I_0/2$ (C) $I_0/4$ (D) Data insufficient

2. In case of linearly polarised light, the magnitude of electric field vector :
(A) Remain same (B) Vary with time linearly (C) Oscillating with time (D) None

3. The resultant amplitude of a vibrating particle by the superposition of the two waves

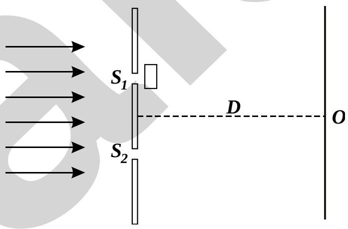
$y_1 = a\sin\left[\omega t + \dfrac{\pi}{3}\right]$ and $y_2 = a\sin\omega t$ is :

(A) a (B) $\sqrt{2}\,a$ (C) $2a$ (D) $\sqrt{3}\,a$

4. When the angle of incidence of an unpolarised light on a material is 60°, the reflected ray is completely polarised. The velocity of the refracted ray inside the material is (in m/s) :

(A) 3×10^8 (B) $\dfrac{3}{\sqrt{2}}\times10^8$ (C) $\sqrt{3}\times10^8$ (D) 0.5×10^8

5. When a perfectly transparent glass slab of refractive index 1.5 is introduced in front of upper slit of a Young's Double Slit Experiment setup as shown in figure below, the intensity at screen center O reduces to half of its earlier value. Minimum thickness of slab would be [Assume that slab does not absorb any energy]:



(A) λ (B) $\lambda/2$ (C) $\lambda/4$ (D) 2λ

6. Unpolarized light of intensity I_0 is incident on surface of a block of glass at Brewster’s angle. In that case, which one of the following statements is true ?

(A) Reflected light is completely polarized with intensity less than $\dfrac{I_0}{2}$

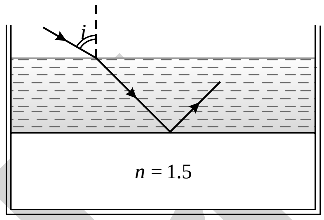
(B) Transmitted light is completely polarized with intensity less than $\dfrac{I_0}{2}$

(C) Transmitted light is partially polarized with intensity $\dfrac{I_0}{2}$

(D) Reflected light is partially polarized with intensity $\dfrac{I_0}{2}$

7. In a double slit experiment, instead of taking slits of equal widths, one slit is made twice as wide as the other. Then in the interference pattern, we have
(A) the intensities of both the maxima and minima increase
(B) the intensity of the maxima increases and the minima has zero intensity
(C) the intensity of the maxima decreases and that of minima increases
(D) the intensity of the maxima decreases and the minima has zero intensity

8. Consider a tank made of glass (refractive index 1.5) with a thick bottom. It is filled with a liquid of refractive index μ. A student finds that, irrespective of what the incident angle i (see figure) is for a beam of light entering the liquid, the light reflected from the liquid glass interface is never completely polarized. For this to happen, the minimum value of μ is :



(A) $\frac{3}{\sqrt{5}}$

(B) $\frac{5}{\sqrt{3}}$

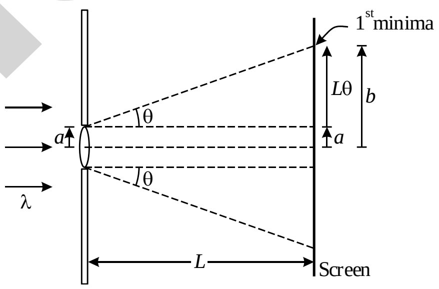
(C) $\sqrt{\frac{5}{3}}$

(D) $\frac{4}{3}$

9. The first diffraction minimum due to the single slit diffraction is seen at $\theta = 30°$ for a light of wavelength 5000 Å falling perpendicularly on the slit. The width of the slit is :
(A) 2.5×10^{-5} cm (B) 1.25×10^{-5} cm (C) 10×10^{-5} cm (D) 5×10^{-5} cm

10. The diameter of the objective lens of microscope makes an angle β at the focus of the microscope. Further, the medium between the object and the lens is an oil of refractive index n. Then the resolving power of the microscope :
(A) Increases with decreasing value of n
(B) Increases with decreasing value of β
(C) Increases with increasing value of $n \sin 2\beta$
(D) Increases with increasing value of $\frac{1}{n \sin 2\beta}$

11. The box of a pin hole camera, of length L, has a hole of radius a. It is assumed that when the hole is illuminated by a parallel beam of light of wavelength λ the spread of the spot (obtained on the opposite wall of the camera) is the sum of its geometrical spread and the spread due to diffraction as specified and shown in figure below. The spot would then have its minimum size (say b_{min}) when :



(A) $a = \frac{\lambda^2}{L}$ and $b_{min} = \left(\frac{2\lambda^2}{L}\right)$
(B) $a = \sqrt{\lambda L}$ and $b_{min} = \left(\frac{2\lambda^2}{L}\right)$
(C) $a = \sqrt{\lambda L}$ and $b_{min} = \sqrt{4\lambda L}$
(D) $a = \frac{\lambda^2}{L}$ and $b_{min} = \sqrt{4\lambda L}$

12. In Young's double slit experiment, the distance between slits and the screen is 1.0 m and monochromatic light of 600 nm is being used. A person standing near the slits is looking at the fringe pattern. When the separation between the slits is varied, the interference pattern disappears for a particular distance d_0 between the slits. If the angular resolution of the eye is $(1/60)°$, the value of d_0 is close to :

(A) 1mm (B) 3mm (C) 2mm (D) 4mm

13. Two stars are 10 light years away from the earth. They are seen through a telescope of objective diameter 30 cm. The wavelength of light is 600 nm. To see the stars just resolved by the telescope, the minimum distance between them should be (1 light year = 9.46×10^{15} m) of the order of :

(A) 10^8 km (B) 10^{10} km (C) 10^{11} km (D) 10^6 km

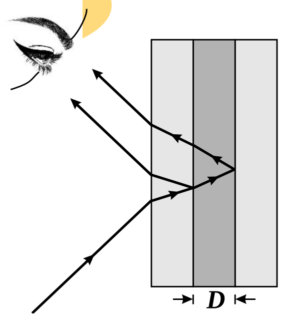
14. Fresnel's fringes are produced with homogeneous light of wave length 6×10^{-5} cm. A thin glass film ($\mu = 1.50$) is introduced in the path of one of the interfering beams. The central bright band is shifted to the position previously occupied by the 5^{th} bright band. Find the thickness of the film :

(A) 3×10^{-4} cm (B) 6×10^{-4} cm (C) 2×10^{-3} cm (D) 4×10^{-3} cm

15. A light of wavelength 5000 Å falls on a glass slab of refractive index 1.5 making angle of refraction 60°. The width of slab so as the reflected rays appear dark is :

(A) 10^{-8} m (B) 4×10^{-7} m (C) $\frac{10}{3} \times 10^{-7}$ m (D) 6×10^{-7} m

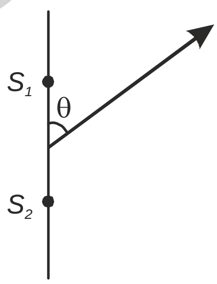
16. Two glass plates, each with an index of refraction of 1.55, are separated by a small distance D. The space between the plates is filled with water (index of refraction of 1.33) as shown. For which one of the following conditions will the reflected light appear green? (Wavelength of green light is 460 nm in vacuum)



(A) $D = \left(\frac{460\text{ nm}}{2}\right)$ (B) $2D = \left(\frac{460\text{ nm}}{1.33}\right)$

(C) $2D = \left(\frac{460\text{ nm}}{1.55}\right)$ (D) $2D = \frac{1}{2}\left(\frac{460\text{ nm}}{1.33}\right)$

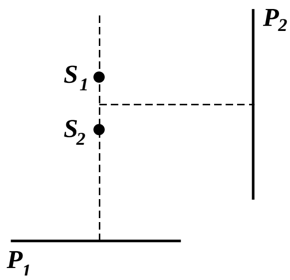
17. In the figure there are two coherent point sources S_1 and S_2 located in a certain plane. They are separated by a distance d. The radiation wavelength is equal to λ. Taking into account that oscillation of source S_2 lag in phase behind the oscillations of source S_1 by ϕ ($\phi < \pi$) then the angle 'θ' at which the radiation intensity can be maximum will be :
(Given : the observation point distance from the centre of S_1S_2 is much greater than d)



(A) $\cos^{-1}\left[\left(n - \frac{\phi}{2\pi}\right)\frac{\lambda}{d}\right]$ $n = 0; \pm 1; \pm 2$ (B) $\sin^{-1}\left[\left(n - \frac{\phi}{2\pi}\right)\frac{\lambda}{d}\right]$ $n = 0; \pm 1; \pm 2$

(C) $\cos^{-1}\left[\left(\frac{n}{2} + \frac{\phi}{2\pi}\right)\frac{\lambda}{d}\right]$ $n = 0; \pm 1; \pm 2$ (D) $\sin^{-1}\left[\left(\frac{n}{2} + \frac{\phi}{2\pi}\right)\frac{\lambda}{d}\right]$ $n = 0; \pm 1; \pm 2$

18. Consider two coherent point sources (S_1 and S_2) separated by a small distance along a vertical line and two screens P_1 and P_2 placed as shown in Figure. Which one of the choices represents the shapes of the interference fringes at the central regions on the screens ?



(A) Circular on P_1 and straight lines on P_2
(B) Circular on P_1 and circular on P_2
(C) Straight lines on P_1 and straight lines on P_2
(D) Straight lines on P_1 and circular on P_2

19. Polarised glass is used in sun glasses because :
(A) It reduces the light intensity to half an account of polarisation
(B) It is fashionable
(C) It has good colour
(D) It is cheaper

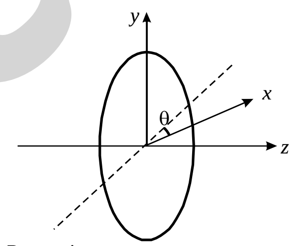
20. Plane polarised light is passed through a polaroid. On viewing through the polaroid we find that when the polaroid is given one complete rotation about the direction of the light, one of the following is observed.
(A) The intensity of light gradually decreases to zero and remains at zero
(B) The intensity of light gradually increases to a maximum and remains at maximum
(C) There is no change in intensity
(D) The intensity of light is twice maximum and twice zero

<u>Section- B: INTEGER Answer Type Questions</u>

21. In Young's Double Slit Experiment, the slit separation is 0.240 mm. When light of wavelength $\lambda = 500$ nm is incident on the slits, calculate the angular position (in degree) of 240^{th} bright fringe on the screen from screen center.

22. Unpolarized light of intensity I passes through an ideal polarizer A. Another identical polarizer B is placed behind A. The intensity of light beyond B is found to be $I/2$. Now another identical polarizer C is placed between A and B. The intensity beyond B is now found to be $I/8$. Calculate the angle (in degree) between polarizer A and C.

23. A plane polarized light is incident on a polarizer with its pass axis making angle θ with x-axis, as shown in the figure. At four different values of θ, $\theta = 8^\circ, 38^\circ, 188^\circ$ and 218°, the observed intensities are same. Calculate the angle (in degree) between the direction of polarization and x-axis.



24. The angular width of the central maximum of a diffraction pattern of a single slit pattern is 60°, the slit width being 1 μm. A double slit experiment is now performed with the same wavelength of light and the same slit widths and it is found that the 10^{th} interference maximum is not formed, but all lower maxima are visible. Calculate the separation (in μm) between the double slits.

25. If I_0 is the intensity of the principle maximum in the single slit diffraction pattern then its intensity becomes nI_0 when the slit width is doubled. Find n.

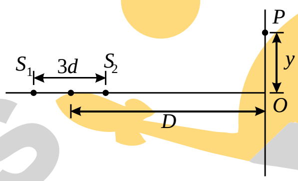
26. In an experiment of single slit diffraction pattern, first minimum for red light coincides with first maximum of some other wavelength. If wavelength of red light is 6600Å, then calculate the other wavelength.

27. A single slit of width 0.1 mm is illuminated by a parallel beam of light of wavelength 6000 Å and diffraction bands are observed on a screen 0.5 m from the slit. Calcualte the distance (in mm) of the third dark band from the central maximum.

28. In a Young's Double Slit Experiment setup, 60 fringes were seen in the field view for light of wavelength 7200 Å. The number of fringes observed when light of wavelength 5400 Å is used, are N. Find N.

29. In Young's Double Slit Experiment setup, if sources are incoherent, the intensity on screen is $13I_0$. When these sources are coherent then minimum intensity on screen is I_0. If maximum intensity produced by these coherent sources on screen is n^2I_0, then find n.

30. Figure shows two coherent microwave source S_1 and S_2 emitting waves of wavelength d and separated by a distance $3d$. For $d << D$ and $y \neq 0$, the minimum value of y for point P to be an intensity maximum is $\frac{\sqrt{m}d}{n}$. Determine the value of $m + n$, if m and n are coprime numbers.



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Topic : WAVE OPTICS ANSWER KEY

Section - A : MCQs with Single Option Correct

1. (B)	**2.** (C)	**3.** (D)	**4.** (C)
5. (B)	**6.** (A)	**7.** (A)	**8.** (A)
9. (C)	**10.** (C)	**11.** (C)	**12.** (C)
13. (A)	**14.** (B)	**15.** (C)	**16.** (D)
17. (A)	**18.** (A)	**19.** (A)	**20.** (D)

Section- B: INTEGER Answer Type Questions

21. [30]	**22.** [45]	**23.** [203]	**24.** [10]
25. [4]	**26.** [4400]	**27.** [9]	**28.** [80]
29. [5]	**30.** [7]		

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